

Supporting Artifacts

Porter: Building First-Time Goods Transport Confidence — the research base, competitive analysis, flows, prototype and measurement system behind the case study.

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01 Research questions & hypotheses

STEP 1

Five questions. RQ1 and RQ3 were formed before collection; RQ2, RQ4 and RQ5 became answerable only once the comparison corpora existed, and are recorded as such rather than presented as predictions that happened to come true.

QUESTION	HYPOTHESIS GOING IN	METHOD	VERDICT
RQ1 What stops a first-time individual customer from completing their first Porter trip, once they have started considering the app?	Pre-booking uncertainty — vehicle size, final price, what is included — causes hesitation and mid-flow abandonment more than post-booking service failures do.	WhatsApp interview, survey free-text, review-mining tags	Supported. Price-opacity and vehicle-fit-uncertainty carry full 3/3 cross-source convergence, joined by packing-material-ambiguity after the second survey wave. Interviewee 1 is a directly observed abandonment.
RQ2 Is Porter's support experience part of first-time conversion, or a separate later-stage retention problem?	Not formed going in — this question emerged from the review data, where support-unresponsive is the largest tag by a wide margin.	Tag frequency cross-checked against survey and interviews	Open. 45/220 of the Porter corpus, more than double any other tag — and zero of 17 survey responses and zero of 4 interviews raised it independently. Deliberately left unresolved rather than forced closed.
RQ3 How does Porter's first-time experience compare with the alternatives a first-timer actually chooses between?	Offline operators win on trust and price predictability rather than price itself; rival apps share Porter's weaknesses rather than solving them.	Competitive synthesis across three corpora plus offline-user responses	Partially supported. R2 and R4 both name negotiation and predictability, not price, as the reason for going offline. Rapido's driver-renegotiation rate is not clearly better than Porter's.
RQ4 Are Porter's barriers goods-transport-specific, or shared with ride-hailing?	Not formed going in — answerable only once Rapido and Uber were collected as comparison corpora.	Tag-by-tag comparison across Porter, Rapido, Uber	Supported. Vehicle-fit-uncertainty and packing-material-ambiguity appear in every goods-transport source and in neither ride-hailing corpus, because neither Rapido nor Uber asks a user to size a load.
RQ5 Does one persona cover the first-time individual customer, or are there meaningfully different sub-types?	Not formed going in — built after RQ1–4, once enough behavioural detail had accumulated.	Synthesis of survey segments, interviews and first-time-signalled reviews	Supported, provisionally. Three types. Later panel waves added nine responses and every one sorted into an existing type — which is itself the useful result.

02

Research design

STEP 1

Three methods — one secondary corpus and two primary instruments — chosen so that each covers a failure the others cannot see.

Method 1 — Review mining at scale

SECONDARY

427 public reviews collected and tagged: **Porter n=220, Rapido n=180, Uber n=27** (India-filtered from 100 collected). Google Play Store, collected 2026-08-28 with a purpose-built scraper.

The corpus is secondary by construction — it was collected, not elicited — and that is precisely where its two advantages come from. The accounts are **unprompted**: nobody was asked about vehicle size, so nobody was led toward it. And they are **uncontaminated by interviewer presence**. The comparison corpora do the work a single-product corpus cannot: they separate a barrier structural to intra-city goods transport from an artifact of one app's interface.

Tagging is conservative. 59 of the 220 Porter rows carry full review text; 161 were tagged from snippets, and 80 of those produced no confident match and are recorded as `untagged-needs-full-text-review` rather than force-fitted into a tag. **Every tag count in this pack is a floor, not a ceiling.**

Method 2 — Survey

PRIMARY

17 responses across two channels: a Google Form distributed in a product-management WhatsApp community (7), and a SurveySwap panel (10). The panel is international by construction, and that is the point — three respondents are first-time users of goods-transport services outside India (U-Haul, Yandex Go, Mudango). **No Porter-specific claim rests on them.** They exist to test whether the primary barrier is a property of the category or of one app. Instrument in §03.

Method 3 — Live interviews

PRIMARY

Four WhatsApp interviews, same opening prompt for each. One (Interviewee 3) was excluded from synthesis: the reply was not a first-person account. It is retained in the log and screenshots as a record of outreach, and cited nowhere.

Sequence

PHASE	WORK	WHAT IT PRODUCED
1 • Collection to 28 Aug	Scraper built and run against the Play Store for Porter, then Rapido and Uber as comparison corpora. Google Form written and distributed to a product-management community; SurveySwap panel opened; interview outreach on WhatsApp.	427 reviews, 8 survey responses, 4 interviews. Enough to see the shape of the problem, not yet enough to separate a Porter issue from a category one.
2 • Tagging 28 Aug	Single taxonomy applied across all three corpora; unmatched rows flagged rather than force-fitted. Cross-source convergence scored per barrier.	The barrier table in §07, and the discovery that convergence and frequency disagree — which became RQ2.
3 • Panel waves 29–30 Aug	SurveySwap panel ran to completion, more than doubling the survey base and adding first-time users of goods-transport services across four further markets.	Survey base to n=17. Vehicle-fit uncertainty corroborated across four markets and four unrelated services; packing-material ambiguity moved from 2/3 to 3/3.
4 • Synthesis 29 Aug	Personas, barrier ranking and competitive analysis rebuilt against the fuller base.	Three personas, and the useful negative result that every new response sorted into an existing one rather than forcing a fourth.

What each method can and cannot see

The steepest leak in this funnel is **structurally invisible to review corpora**: someone who abandons before completing a booking has no reason to write an app-store review, and a successful vehicle fit generates no review either. Reviews over-represent post-booking service failures and under-represent configuration failure. The interviews and survey exist to reach exactly that blind spot — and Interviewee 1 is the one directly observed abandonment in the whole base.

03 Respondent & interview log

STEP 1

Every human research interaction in the base, with segment and the insight it contributed.

Survey respondents and WhatsApp interviewees are anonymised in this public version of the pack.

ID	CHANNEL	IDENTIFIER	SEGMENT	CONTRIBUTION
R1	Google Form	—	Hypothetical / non-user	Scenario response; no first-hand booking experience.
R2	Google Form	—	Used offline — professional packers & movers	Core offline-loyalist evidence: “if I had more information about the reliability and safety of the app-based services, I might have considered trying one”; also “needed help with loading/unloading, not just driving.”
R3	Google Form	—	Hypothetical / non-user	Scenario response.
R4	Google Form	—	Used offline — local operator / own arrangement	Chose own arrangement to avoid driver-reliability uncertainty; names negotiation as the reason for going offline.
R5	Google Form	—	First-time app-based user — on target	Load-bearing. Five simultaneous pre-booking unknowns: “Which vehicle size I needed; Whether my stuff would actually fit; What the total cost would be; How long it would take; Whether my things would get damaged.” Also reports “vehicle was too small / wrong type.”
R6	Google Form	—	Hypothetical / non-user	No substantive content.
R7	Google Form	—	Hypothetical / non-user	Scenario response.
R8	SurveySwap	A	First-time app-based user — on target	Load-bearing. Unprompted feature request: “scanning my things and telling me approx which size vehicle would I need.” Also “final price differed from the estimate.”
R9	SurveySwap	Not disclosed	First-time U-Haul user (US)	Cross-market corroboration: “Which vehicle size I needed; Whether my stuff would actually fit” — and “didn’t trust an app to send the right vehicle” as the reason to go offline.
R10	SurveySwap	Aurora	First-time app-based user	Sorted into The Cautious First-Timer; no new barrier introduced.
R11	SurveySwap	eha	First-time app-based user	Sorted into The Cautious First-Timer.
R12	SurveySwap	Mary	First-time Yandex Go user (Russia/CIS)	Cross-market corroboration: “How much it would cost and whether the vehicle would be suitable for the item”; wants “a guarantee that the right vehicle would be provided.”
R13	SurveySwap	Alvaro Diaz	First-time Mudango user (Chile)	Moved packing-material-ambiguity to 3/3 convergence: names “whether the driver could help load/unload” as a pre-booking unknown.
R14	SurveySwap	Lojain	Prior app user — no substantive answers	Recorded for completeness; not used in synthesis.
R15	SurveySwap	KG	First-time Porter user — on target	Load-bearing. The second Porter first-timer to name “Which vehicle size I needed” and “Whether my stuff would actually fit” among their pre-booking unknowns and then report “vehicle was too small / wrong type” as what happened, alongside a price difference and damage. Confidence 2 of 5 before booking. Now prefers offline, because operators are “transparent with the cost, vehicle size and load.”
R16	SurveySwap	Anastasiia Budanova	First-time — used local contacts	Names “Which vehicle size I needed” and “Whether my stuff would actually fit” as pre-booking unknowns. Asked what she would want to know first before booking a fridge or sofa, answers “if vehicles big enough.” Went offline partly because she “didn’t know an app option existed for this.”
R17	SurveySwap	Zen Ooi	No prior booking experience	Recorded for completeness; not used in synthesis.

ID	CHANNEL	IDENTIFIER	SEGMENT	CONTRIBUTION
I1	WhatsApp interview	Interviewee 1	Considered Porter, abandoned the booking	The single most load-bearing account in the base. Abandoned a car-seat booking over unclear packing-material policy and an unexpected loading charge; also felt forced into a larger, costlier vehicle. <i>“Eventually I ended up not using the service.”</i>
I2	WhatsApp interview	Interviewee 2	Repeat user — Porter and Uber/Rapido	Competitive and trust evidence: <i>“I’ll get a call from courier person asking me to pay extra offline since 150 is too less.”</i> Segment mismatch noted — repeat, not first-time.
I3	WhatsApp interview	Interviewee 3	—	Excluded from synthesis. The reply was not a first-person account. Logged as evidence of outreach; cited nowhere in this case study.
I4	WhatsApp interview	Interviewee 4 ANONYMISED FOR WEB	Not established in the exchange	Wants pre-booking route and driver visibility: <i>“I would want truck driving information before booking it and also 4-5 different route option available in the app.”</i>
—	Distribution record	PM community group	Recruitment channel	Confirms the Google Form was posted to a real product-management community and completed by members. Screenshot in \$04.

21 direct human interactions — 17 survey responses and 4 interviews — alongside the 427-review corpus. Interviewee 3 is counted as an interaction and excluded from every finding.

Survey instrument

Identical across both channels. Fourteen substantive questions, sequenced so that unaided recall of pre-booking uncertainty is captured before any question names a candidate barrier.

PURPOSE	QUESTIONS
SCREENER	Have you needed to move furniture, appliances, boxes or other bulky items across your city in the last 12 months — not a full home relocation? · Which service did you use? · Was this your first time using an app-based service for this kind of job?
PRE-BOOKING STATE	Right before you booked, what were you unsure about? (<i>multi-select</i>) · Right before you tapped “book,” how confident were you the trip would go as planned?
OUTCOME	What actually happened? (<i>multi-select</i>) · If you had to do it again tomorrow, would you book the same way, and why?
NON-ADOPTERS	Why did you go with that option instead of an app-based service? (<i>multi-select</i>) · What would have made you try an app-based service instead?
UNAIDED PRIORITY	If you needed to move a fridge or sofa across your city tomorrow, what is the first thing you would want to know before booking anything?
CHANNEL CHOICE	Would you be more likely to try an app-based service or a local/offline contact first? · Why?
OPEN	Anything about moving bulky goods within a city that surprised you, worried you, or you wish was easier?
RECRUITMENT	Open to a 10-minute follow-up chat about this?

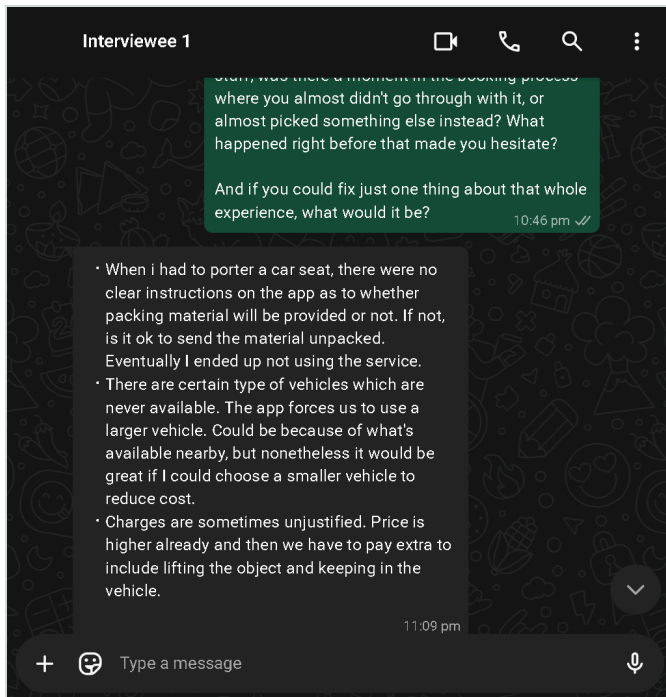
Why the unaided-priority question carries the most weight

It asks what a first-timer wants to know before booking without naming vehicles, price, timing or any other candidate barrier. A sizing answer there is unprompted, and it is the only question in the instrument that can corroborate the review corpus without inheriting its framing. R8’s request for a tool that scans “my things” and returns a vehicle size came from this question.

04 Conversation evidence

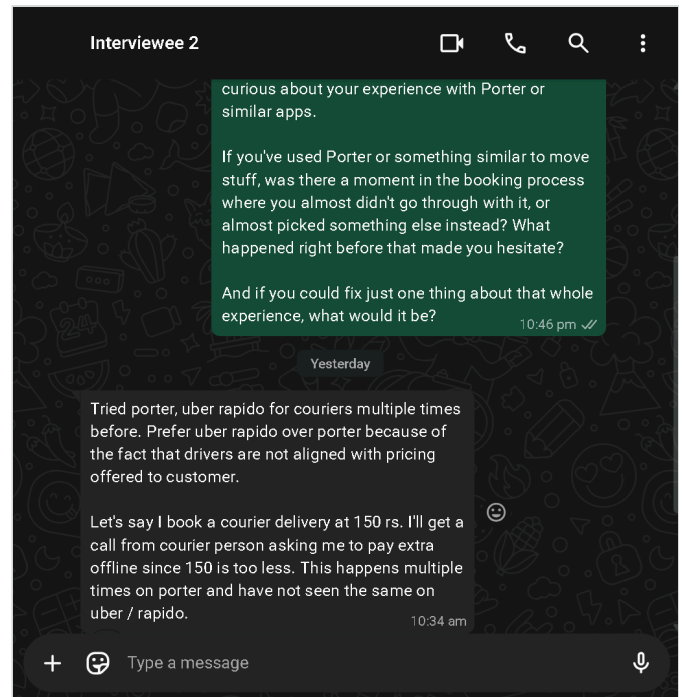
STEP 1

Verbatim exchanges, redacted. Profile photos and contact names are masked; the interview prompt is visible in each so the questions can be judged alongside the answers.



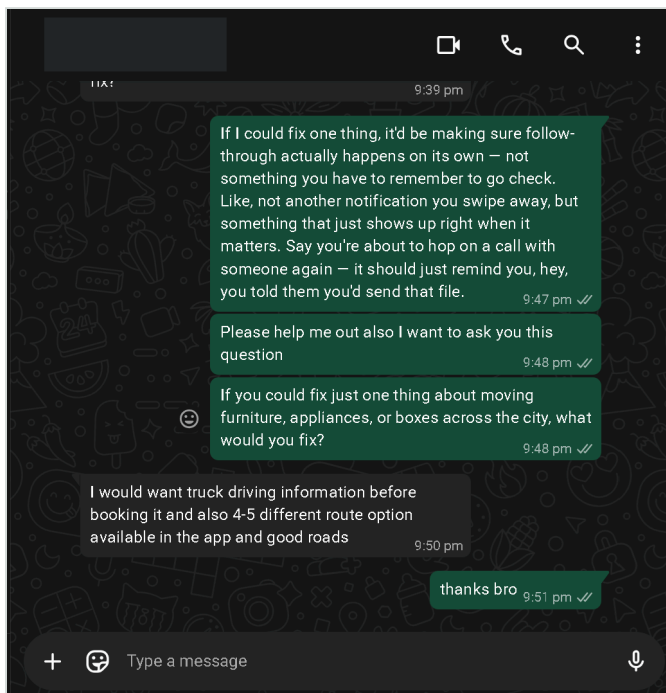
Interviewee 1 — the abandoned booking

Three barriers in one reply: no clear packing-material policy, forced into a larger vehicle than needed, and unjustified extra charges for lifting. Ends with the line the whole case study turns on — *"Eventually I ended up not using the service."*



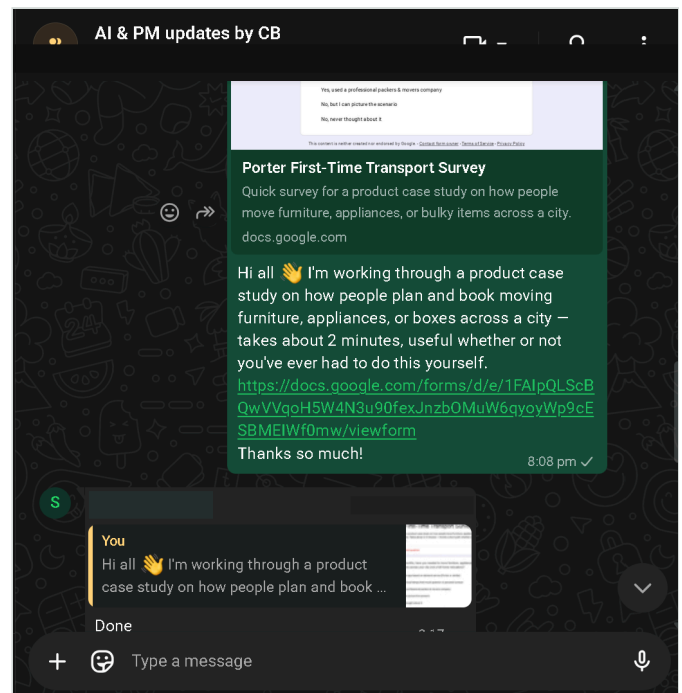
Interviewee 2 — driver-side renegotiation

Books at ₹150 and is called for extra cash off-app, repeatedly. Prefers Uber/Rapido as a result. Repeat user rather than first-time, so used as competitive and trust evidence only.



Interviewee 4 — pre-booking visibility

Asks for driver and route information *before* booking rather than after. Contributes the tracking-visibility barrier. Name and photo masked.



Distribution record

The Google Form posted to a product-management community, with a member confirming completion. Establishes that the survey channel was a real community rather than a personal contact list. Member identifiers masked.

05

Review-mining corpus

STEP 1

427 reviews collected 2026-08-28 from the Google Play Store and tagged against a single taxonomy, so that Porter's barriers can be compared with two ride-hailing corpora rather than read in isolation.

Porter tag frequency — n=220

TAG	COUNT	%
untagged-needs-full-text-review	80	36%
support-unresponsive	45	20%
price-opacity	22	10%
trust-unfamiliarity	19	9%
fraud-safety-concern	19	9%
driver-no-show-cancellation	14	6%
positive-signal	14	6%
tracking-visibility	13	6%
app-bug	13	6%
damage-handling	11	5%
scheduling-availability	9	4%
driver-side-renegotiation	9	4%
packing-material-ambiguity	6	3%
account-punitive-action	6	3%
booking-abandonment	5	2%
vehicle-fit-uncertainty	5	2%

Star distribution — n=220

STARS	COUNT	%
5	18	8%
4	9	4%
3	8	4%
2	11	5%
1	174	79%

Corpus confidence by source

PORTER	Highest. n=220; 59 rows full-text tagged, 80 honestly flagged as unmatched. Primary source.
RAPIDO	Medium-low. n=180 but 68% untagged and almost none of it about goods transport. Read as a platform-trust signal only.
UBER	Lowest. n=27 after India-filtering 100; only 4 carry an explicit India marker. No single number from this column stands alone.
OFFLINE	Thinnest. Two survey respondents. Real evidence; not a base rate.

Two things this table cannot be read as

The 79% one-star skew is a sampling artifact, not a satisfaction measure. Play Store's default sort surfaces critical reviews first, and Rapido shows a near-identical 73% skew under the same collection method. Read the skew as a property of the collection, not of Porter's customers.

vehicle-fit-uncertainty at 2% is the corpus's blind spot, not a small barrier. This is the lowest corpus percentage of any 3/3-convergence barrier in the whole base, and that is exactly what the method predicts: a user who could not size their load and left never wrote a review, and a user whose vehicle fit perfectly had nothing to complain about. The primary research is far more convinced of this barrier than the corpus is, and the discrepancy is the finding.

06 Competitive analysis

STEP 1

Built from this project's own corpora rather than from secondary sources. The final column records where the comparison does not hold — those tensions matter more than the cells that agree.

DIMENSION	PORTER	RAPIDO	UBER	OFFLINE OPERATOR	TENSION
Price transparency	Weak. price-opacity 22/220 (10%); corroborated by R8 and Interviewee 1.	Weak, same shape. 5/180 tagged (3%), likely undercounted at 68% untagged.	Same pattern in n=27; too small to weight.	Reported as a strength. R2: <i>"price felt more predictable."</i>	Likely category-wide rather than Porter-specific. Check before positioning pricing as a differentiator.
Driver-side renegotiation	9/220 (4%).	5/180 (3%) — not obviously better.	1/27; too small to read.	Not applicable — the price is the negotiation.	Interviewee 2 switched to Rapido over exactly this, but Rapido's own rate is close to Porter's. The account may not generalise.
Support responsiveness	Porter's largest tagged issue: 45/220 (20%).	16/180 (9%), likely undercounted.	Most-cited tag in the small set too.	Strength by construction — support is the same person who quoted the price.	Porter's most corroborated weakness across every source that has data, and the strongest candidate for a Porter-specific rather than category-wide problem.
Damage handling	damage-handling 11/220 (5%); fraud-safety 19/220 (9%).	2/180 (1%) — but almost no Rapido reviews concern goods.	Insufficient data.	No review evidence either way.	Not a meaningful cross-app comparison. Do not cite this row as Porter being worse.
First-time trust	19/220 (9%); R5 unsure of size, fit, cost, damage and duration at once.	11/180 (6%).	1/27.	The offline option's core appeal (R2).	The strongest cross-source convergence in the file — survey, interview, and all three corpora.
Scheduling / no-shows	scheduling 9/220 (4%); no-show 14/220 (6%).	4/180 and 1/180.	1/27 each.	R4 chose their own vehicle to avoid this entirely.	Porter is highest of the three apps here, though Rapido's true rate is uncertain.
Vehicle fit & packing clarity	vehicle-fit 5/220 (2%); packing 6/220 (3%). Interviewee 1 abandoned outright over both.	N/A — no comparable sizing decision.	N/A — same reason.	Structural strength. A human operator sizes the job before committing — precisely the missing step.	Porter's most goods-transport-specific weakness, not shared with ride-hailing by product design, and the one with an actual abandoned booking behind it.

The competitor that matters most is not an app

For the segment that never enters Porter's funnel, the real alternative is a phone call. Both offline respondents chose it for the same two reasons — the ability to negotiate, and the ability to talk to a person who sizes the job before committing. Neither reason is price. That last capability is the one Porter's flow does not have, and it is what the solution in this case study is built to replace.

07 Top anticipated barriers

STEP 1 → STEP 2

Ranked by **convergence** — how many independent source types name the barrier — rather than by review-corpus frequency. The two lenses disagree, and where they disagree is informative.

BARRIER	CONV.	PORTER CORPUS	SURVEY / INTERVIEW EVIDENCE	NOTE
vehicle-fit-uncertainty	3/3	5/220 (2%)	R5, R9, R12, R15 and R16 name vehicle size and fit as pre-booking unknowns across four markets. Interviewee 1 pushed into a larger vehicle. R8 requests a scan-based sizing tool unprompted.	Selected as the primary barrier. Lowest corpus % of any 3/3 barrier — the corpus structurally cannot see it. Absent from both ride-hailing corpora, which have no sizing decision.
price-opacity	3/3	22/220 (10%)	R8: “final price differed from the estimate.” R12 wants clear pricing. R15 reports the same gap. Interviewee 1: unexpected loading charge.	Strong, but category-wide per RQ4, and a fix grazes the out-of-scope pricing-engine line.
packing-material-ambiguity	3/3	6/220 (3%)	R13: “whether the driver could help load/unload.” R2 from the offline side. Interviewee 1’s loading charge.	Upgraded from 2/3 by the second survey wave. Sits in the same pre-booking moment as vehicle fit and is not answered by a sizing recommendation.
trust-unfamiliarity	2/3	19/220 (9%)	R2 would try an app with more reliability information. R5 unsure of everything at once.	Users fuse this with vehicle fit: R2 and R9 both give “didn’t trust an app to send the right vehicle” as the reason to go offline.
damage-handling	2/3	11/220 (5%)	R15 reports “items got damaged” as an outcome. R5 lists it as a pre-booking worry.	Moved from 1/3 when a survey respondent reported it as an outcome rather than a worry. Downstream of vehicle fit: the wrong vehicle is how goods get damaged.
tracking-visibility	2/3	13/220 (6%)	Interviewee 4 wants driver and route visibility before booking.	Pre-booking rather than in-transit framing.
driver-side-renegotiation	2/3	9/220 (4%)	Interviewee 2 switched apps over it.	Rapido’s rate is not clearly lower — see \$06.
booking-abandonment	2/3	5/220 (2%)	Interviewee 1 — the one directly observed case.	Under-counted everywhere by construction: abandoners do not review.
support-unresponsive	1/3	45/220 (20%)	Not raised by any of the 17 survey responses or 4 interviews.	The sharpest frequency-versus-convergence disagreement in the file. Left open as RQ2.
fraud-safety-concern	1/3	19/220 (9%)	Adjacent only, via Interviewee 2.	Reads as unwanted upsell rather than fraud; not counted.
driver-no-show-cancellation	1/3	14/220 (6%)	Not raised.	Corpus-only.
app-bug	1/3	13/220 (6%)	Not raised.	Corpus-only.
scheduling-availability	1/3	9/220 (4%)	R4 adjacent only.	Not counted as a hit.
account-punitive-action	1/3	6/220 (3%)	Not raised.	Corpus-only.

Convergence counts source *types* (survey / interview / corpus), not mentions. Interviewee 1’s abandonment names three causes and is counted once against each — one event, three barriers, not four independent witnesses.

08 First-time transport personas

STEP 1

Three types, built from the response data rather than assumed in advance. The second survey wave added six responses and every one sorted into an existing type without forcing a fourth.

1 • The Cautious First-Timer

Wants to try Porter, but arrives with more open questions than the booking flow answers.

EVIDENCE	R5, R8 (Porter), R9 (U-Haul), R12 (Yandex Go), R13 (Mudango); first-time-signalled rows in the review corpus.
FUNNEL STAGE	Pre-booking hesitation through to a completed-but-rocky first trip. Closest of the three to the case study's conversion metric.
WANTS	To know before tapping book: what vehicle size, whether the items fit, the real total cost, how long, and whether anything gets damaged.
BARRIERS	vehicle-fit-uncertainty, price-opacity, trust-unfamiliarity, packing-material-ambiguity, damage-handling.
QUOTE	"Which vehicle size I needed; Whether my stuff would actually fit; What the total cost would be; How long it would take; Whether my things would get damaged" — R5, all five given as pre-booking unknowns in one answer.
NOT THIS	Not a repeat user weighing Porter against a habit. This is someone giving Porter a genuine first attempt.

2 • The Abandoner

Opens the app, starts a booking, and walks away before the trip happens.

EVIDENCE	Interviewee 1 — considered Porter for a car-seat move, started, did not complete. Corroborated in the corpus by booking-abandonment rows.
FUNNEL STAGE	Mid-funnel abandonment — the exact failure the case study defines success against.
WANTS	A straightforward, predictable first move: know the packing responsibility, the real price, and the right vehicle before committing.
BARRIERS	packing-material-ambiguity, price-opacity, vehicle-fit-uncertainty, booking-abandonment — all four converge on one account.
QUOTE	"Eventually I ended up not using the service."
NOT THIS	Not someone who never opened the app, and not someone whose trip completed roughly. Specifically the drop-off-mid-flow case.

3 • The Offline Loyalist

Has bulky items to move, knows apps exist, and calls a person instead — every time.

EVIDENCE	R2 (professional packers & movers) and R4 (local operator / own arrangement). Still n=2; the second wave added none.
FUNNEL STAGE	Active avoidance — has not entered Porter's funnel at all. The competitive set here is a phone call, not Rapido.
WANTS	To negotiate directly, talk to a person before committing, and trust the job will be done right. Predictability over convenience.
BARRIERS	trust-unfamiliarity is the named reason; price-opacity is what direct negotiation avoids.
QUOTE	"If I had more information about the reliability and safety of the app-based services, I might have considered trying one."
NOT THIS	Not a first-time Porter user who tried and bounced. This segment has not tried an app for this job at all.

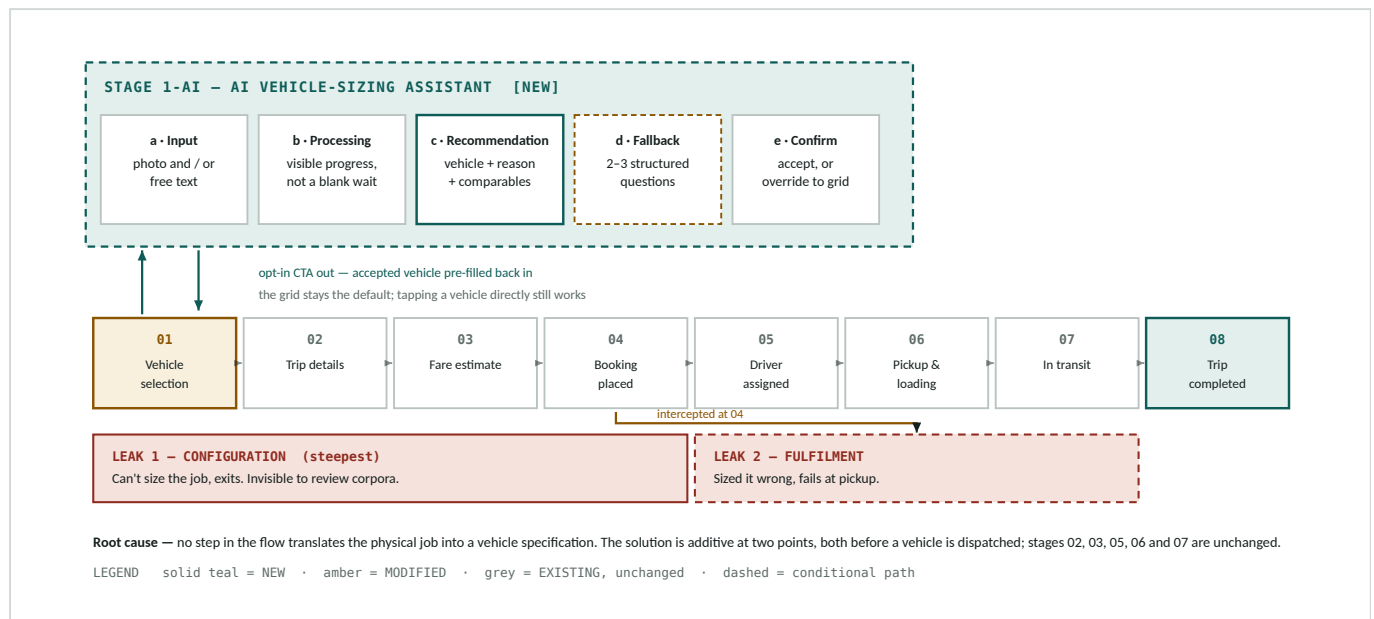
09 Problem, funnel & business impact

STEPS 2 & 5

First-time individual customers reach Porter's vehicle-selection screen with a concrete physical job — a refrigerator, a sofa, a car seat, a room's worth of cartons — and are asked to convert it into an abstract vehicle category. Nothing in the app tells them which one fits. Repeat and business users have built that judgement across dozens of trips; first-timers guess. The guess is where the funnel breaks: some cannot make it and leave, and some make it wrong and find out when the vehicle arrives. Both are the same failure, because Porter counts a first trip as successful only when the trip completes — not when the booking is placed.

REFINED PROBLEM STATEMENT

The denominator opens at vehicle selection — everything upstream is outside the metric. Two leaks, one root cause, and two additive touchpoints — both before a vehicle is dispatched: **no step in the flow translates the physical job into a vehicle specification.**



Why the fulfilment protection sits at Stage 04 and not at pickup

Two versions of this check were designed at the pickup stage first — a customer-initiated capacity flag, then a driver-initiated one with a one-tap fare approval. Both lost to the same thing: at the curb, with the load half-carried, a driver naming a cash number is faster than any in-app flow. The driver-initiated version additionally depended on a driver choosing to log a mismatch they profit more from settling informally, and driver compensation is out of scope to change.

Moving the same protection to Stage 04 removes every one of those objections at once. **There is no driver present to compete with, and no dispatch to unwind** — the booking has not been created, so a switch is a re-selection rather than a re-assignment, which keeps the design clear of the out-of-scope backend-dispatch line. The cost of being wrong becomes a fare difference the user chooses rather than a negotiation they are ambushed with.

It buys prevention, not recovery. A load described inaccurately and sized confidently passes the triggers unflagged; the Stage 08 fit confirmation and metric F3 in §14 size that residual rather than assume it away.

Business impact

41 / 58 AND THE 10 LAKH INDIVIDUAL BASE: CASE STUDY

VOLUMES, AOV, CAC, REPEAT RATE: ILLUSTRATIVE ASSUMPTIONS

Assumptions carried consistently throughout: 1.5 lakh first-timers reaching vehicle selection per month (~15% of the individual base), AOV ₹800, CAC ₹250, repeat rate 30%, 2.5 further trips over 24 months.

VOLUME – FIRST-TIME INDIVIDUAL CUSTOMERS			
LINE	TODAY – 41%	TARGET – 58%	Δ
Completed first trips / month	61,500	87,000	+25,500
Completed first trips / year	7.38 L	10.44 L	+3.06 L
First-timers lost / year	10.62 L	7.56 L	–3.06 L

LIFETIME VALUE – 24 MONTHS PER FIRST-TIME CUSTOMER			
PER FIRST-TIMER REACHING THE VEHICLE-SELECTION SCREEN	TODAY	TARGET	Δ
Value if the first trip completes — ₹800 + (0.30 × 2.5 × ₹800)	₹1,400	₹1,400	–
Expected 24-month revenue	₹574	₹812	+₹238
Less CAC — spent regardless of outcome	–₹250	–₹250	–
Net expected value per first-timer	₹324	₹562	+73%
Across the annual cohort — 18 lakh first-timers	₹58.3 Cr	₹101.2 Cr	+₹42.8 Cr

The value of a completed customer does not change. The feature does not make a customer worth more — it makes more customers complete. So the whole effect lands on the expected value of a first-timer *entering* the funnel, and the leverage is larger than a 17-point move looks: the entire ₹250 of acquisition cost currently rides on a coin-flip, and this improves the flip.

Opportunity size

LAYER	WHO	SIZE	BASIS
TAM	Every Porter intra-city customer, individual and MSME	30 lakh / month	Case study
SAM	Individual (non-MSME) customers. MSME is out of scope and buys on different criteria.	10 lakh / month	Case study
SOM	First-time individuals reaching vehicle selection — the funnel stage this solution acts on	1.5 lakh / month 18 lakh / year	~15% first-timer share; illustrative

TAM and SAM are context. SOM is the layer this solution is accountable for, and its addressable value is the ₹42.8 Cr a year derived above.

One reconciliation line, because these numbers can be added wrongly

The +₹42.8 Cr is the same money as the ₹43 Cr incremental GMV per annual cohort reached the other way — ₹24.5 Cr of first-trip revenue plus ₹18.4 Cr of downstream repeat. Two routes, one number. A third line, ₹7.7 Cr of acquisition spend recovered, is **not additive to it**: CAC is spent at acquisition whether or not the trip completes, so it cancels out of the delta entirely. Anyone summing all three reaches ₹50.5 Cr, and that number is wrong.

SENSITIVITY – INCREMENTAL VALUE PER ANNUAL COHORT			
INPUT VARIED	LOW	BASE	HIGH
First-timer share of the 10 lakh individual base — the softest input	10% · ₹28.6 Cr	15% · ₹42.8 Cr	20% · ₹57.1 Cr
Average order value	₹600 · ₹32.1 Cr	₹800 · ₹42.8 Cr	₹1,000 · ₹53.6 Cr

Repeat-rate lift is deliberately *not* in the base case. A move from 30% to 35% would add ₹10.4 Cr per annual cohort and applies to the whole cohort rather than the incremental slice — but that is an argument, not a measurement, and building it in would score the model in the solution's own favour. Guardrail G4 in §14 is the instrument pointed at it.

10 Prioritization — RICE

STEP 3

Ten solutions, scored against all barriers rather than only the primary one. **Reach** = first-time individual customers exposed to the trigger moment per month, in absolute users. **Impact** tied to convergence. **Confidence** mapped from convergence (3/3 = 80%, 2/3 = 50%, 1/3 = 25%). **Effort** = real Porter engineering cost to ship in production — explicitly not how easy an idea is to mock up for a submission.

Two reach populations. 1.50 L first-timers reach vehicle selection each month (\$09). Interventions that fire after a booking is placed reach fewer: at an illustrative Stage 1→4 pass-through of 55%, that is **0.82 L** bookings and **0.68 L** mid-flow exits. The split reconciles to the case study's own figure — $55\% \times 75\% = 41\%$ completion — and keeps Leak 1 the steeper of the two.

#	SOLUTION	BARRIER	LEAK	REACH / MO	I	C	E	RICE
1	AI vehicle-sizing assistant — photo or text input, recommends a vehicle, surfaces comparable past bookings	vehicle-fit + trust	1	1.50 L	3	80%	L=4	0.90
2	Guided item-configurator wizard — non-AI, rules-based lookup	vehicle-fit	1	1.50 L	2	80%	S=1	2.40
3	Post-selection capacity confirm + in-app upgrade before dispatch	vehicle-fit	2	0.82 L	2	80%	M=2	0.66
4	Itemized upfront fare breakdown + conditional-cost disclosure	price-opacity	1	1.50 L	2	80%	S=1	2.40
5	Pre-booking orientation prompt — packing responsibility, documents, protection	packing + damage	1	1.50 L	2	50%	S=1	1.50
6	First-time orientation content at the entry point, not buried in FAQ	trust	1	1.50 L	1	50%	S=1	0.75
7	Pre-booking route and driver-process transparency	tracking	1	1.50 L	1	50%	M=2	0.38
8	Save-and-resume + automated re-engagement on mid-flow exit	abandonment	1	0.68 L	2	50%	M=2	0.34
9	Fare-lock guarantee + in-app dispute flag	renegotiation	2	0.82 L	1	50%	M=2	0.21
10	Proactive trip-status updates, pickup to delivery	support (RQ2 open)	2	0.82 L	0.5	25%	M=2	0.05

Top 2 — and why they are not the top-RICE pair

#1 AI vehicle-sizing assistant — primary, built. The only idea scoring Impact = 3; the most direct answer to the primary barrier; grounded in R8's unprompted request for exactly this capability.

#2 Guided item-configurator wizard — runner-up, argued not built. Tied-highest RICE, and it answers the same barrier from the opposite direction — deterministic and instant versus AI-guided and rich. That opposition is what makes the required guidance-versus-speed trade-off concrete instead of generic.

#4 ties on RICE but answers a different barrier; the top two were kept aligned with the primary-barrier reasoning rather than diversified. **#3 was not discarded either:** its *before-dispatch* form is folded into the primary solution as the conditional Stage 04 load check (\$09), which is the placement its RICE row always described — a pickup-stage version was designed later, tested against the curb, and rejected. Two of the researcher's own candidate ideas were evaluated and dropped for cause: a peer-advisor call network (bordered on the banned manual-concierge lever and required sharing customer contact details) and a resale-shipping API partnership (different user, different problem, channel outside scope).

Score justification

One line per idea, so every number above can be checked rather than taken.

#	WHY THOSE NUMBERS
1	R every first-timer at vehicle selection. I=3 the only idea that answers the primary barrier directly and the only one carrying a trust layer alongside the recommendation. C=80% 3/3 convergence. E=L vision model, item taxonomy, fallback rules table, four new tracking events.
2	R same screen, same moment. I=2 answers sizing but carries no trust layer and breaks on items outside its fixed taxonomy. C=80% same convergence. E=S a rules table and a form.
3	R fires only after a booking exists, so bookings-placed rather than the full cohort. I=2 catches mis-sizing but only after the user has already chosen wrong. C=80% same barrier evidence. E=M a conditional screen plus re-selection logic.
4	R the fare estimate is on the path for every first-timer. I=2 answers price-opacity, a 3/3 barrier, but not the primary one. C=80% 22/220 corpus plus survey and interview support. E=S presentation of figures the engine already computes.
5	R shown pre-booking to the whole cohort. I=2 packing-material ambiguity is 3/3 but sits adjacent to the primary barrier rather than on it. C=50% upgraded to 3/3 only in the second survey wave, so treated as 2/3-grade evidence. E=S content and placement.
6	R entry-point content reaches everyone. I=1 orientation content moves trust weakly and moves sizing not at all. C=50% 2/3 convergence on trust-unfamiliarity. E=S content only.
7	R pre-booking screens reach the full cohort. I=1 tracking-visibility is 2/3 and framed pre-booking rather than in-transit; it does not touch the completion decision. C=50% 2/3. E=M needs driver and routing data surfaced pre-booking.
8	R only users who exit mid-flow, which is the smallest population of the ten. I=2 recovers an abandoned session but does nothing about why it was abandoned. C=50% abandonment is inferred from the funnel, not named by users — the corpus structurally cannot see it. E=M state persistence plus a re-engagement trigger.
9	R bookings placed. I=1 renegotiation is real but driver-side, and the fix cannot change driver compensation. C=50% 2/3, carried mainly by Interviewee 2, who is a repeat user. E=M fare-lock logic touches the pricing boundary.
10	R dispatched trips only. I=0.5 status updates address a support symptom rather than a booking or fulfilment cause. C=25% 1/3 — RQ2 is deliberately open on whether support responsiveness is a conversion or a retention problem. E=M a notification pipeline against live trip state.

Trade-off analysis

The AI assistant optimizes for recommendation quality and coverage — it handles non-standard loads — and reaches two barriers at once, sizing and trust, on a single screen. It sacrifices speed, since it adds an open-ended interaction before the user can proceed, and predictability, since a model has failure modes a lookup table cannot have. It is the most expensive of the ten to build and maintain.

The configurator wizard optimizes for speed, predictability and cheap engineering — deterministic, testable, fast to ship. It sacrifices precision at the edges: an item outside its fixed taxonomy reproduces the exact ambiguity it exists to remove. It is also narrower, answering sizing with no trust layer.

The shared trade-off, tied to the case study's own constraint that the experience stay quick enough for an on-demand service: *both* buy certainty by adding a step. The wizard costs less time per user than the assistant. Neither is free, and §14's metric B6 is what measures the bill.

One cost the RICE table understates

The assistant's low-confidence fallback reuses the wizard's deterministic rules. The wizard is therefore not avoided by declining to ship it — its rules table still gets built and maintained inside the more expensive AI build. Four new tracking events (§14) and the Stage 04 load check are also work this solution creates rather than inherits. All of it understates idea #1's Effort score, and all of it is recorded rather than quietly dropped.

11 Product requirements — PRD

STEP 4

AI Vehicle-Sizing Assistant, with a conditional load check before dispatch. Owner scope: Porter, first-time individual customer experience. The problem this answers is stated in §07 and mapped in §09; this section specifies what gets built and the boundary around it.

Users

IN SCOPE	First-time individual customers — operationally, an individual (non-MSME) account with zero completed Porter trips. Roughly 10 lakh monthly individual customers is the addressable population this screen serves, derived from the case study's 30 lakh monthly less 20 lakh MSME.
OUT — REPEAT	Repeat individual customers have already built vehicle-sizing judgement across past trips. The tool costs them a step for no benefit, which is why it is opt-in rather than gating the screen.
OUT — BUSINESS	Business and MSME accounts: different booking patterns, often standardised loads, and outside the case study's own boundary.

Success metric

Vehicle-selection → trip-completion conversion for first-time individual customers. **Baseline 41%, target 58%** — both given by the case study, neither modelled. Entry event at Stage 01, completion at Stage 08; the metric is the pass-through between them. Full measurement system, including fulfilment indicators and guardrails, in §14.

Requirements

§	REQUIREMENT	SPECIFICATION
5.1	Entry point STAGE 01, MODIFIED	A CTA — “ <i>Not sure what size you need?</i> ” — on the existing vehicle-selection screen, above or beside the category grid. The grid is unchanged and remains the default path. Prominent for accounts flagged first-time; visible but not pushed for everyone else.
5.2	Input NEW	A single chat-style bar rather than separate modes: a “+” attach control on the left (camera/gallery, multi-image) beside a free-text field, submit on the right. Photos and text can be given together or independently; both converge on the same recommendation call.
5.3	Processing NEW	A loading state with named progress — “ <i>Matching your items to similar past bookings</i> ” — not a blank wait.
5.4	Recommendation NEW	One screen carrying three things together, not tabbed apart: the recommended category stated plainly; a one-line reason tied to the specific input (“ <i>a Canter 14ft comfortably fits a 2-seater sofa, a fridge and ~15 boxes — a Pickup 8ft runs tight on this combination</i> ”); and 2–3 comparable past bookings with a similar load in the same category. The third element answers trust-unfamiliarity, the second barrier this idea was scored against — users fuse the two, giving “ <i>didn't trust an app to send the right vehicle</i> ” as one reason, which is why they are answered on one screen.
5.5	Low-confidence fallback NEW	When the load cannot be classified confidently, the system does not guess silently. It drops into 2–3 structured questions — item count, largest single item, any oversized or irregular pieces — before recommending. No numeric confidence threshold is specified: that is a Porter ML decision, not a number to invent here.
5.6	Confirm or override NEW	Accepting pre-fills the vehicle field at Stage 01 and continues into the existing flow unchanged. Override is always available and never blocked — it opens the full manual grid.
5.7	Load check STAGE 04, CONDITIONAL	One confirmation immediately before the booking is created, on three triggers only: a low-confidence recommendation accepted anyway, a downward override into a smaller category, or a session that used the fallback. It replays the described load against the chosen vehicle and offers a switch, which re-runs the existing fare estimate. There is no driver yet and no dispatch to unwind, so a switch is a re-selection rather than a re-assignment.

§	REQUIREMENT	SPECIFICATION
5.8	Post-trip confirmation STAGE 08, NEW	A one-tap prompt after a completed trip: what was moved, and whether the vehicle fitted — <i>fit / tight / didn't fit</i> . One instrumentation cost, two payoffs: it builds the comparable-moves corpus §5.4 surfaces, and it is the fit signal the customer — unlike the driver — has every reason to give.

Scope boundaries

IN	The assistant as specified in 5.1–5.6 for first-time individual customers; the Stage 01 CTA and its placement logic; the low-confidence fallback; the Stage 04 load check; the Stage 08 post-trip confirmation.
OUT – INHERITED	Driver-partner acquisition, fleet expansion, driver compensation, Porter Enterprise, full Packers & Movers redesign, warehouse operations, interstate logistics, core pricing-engine redesign, insurance-policy redesign, vehicle procurement, backend dispatch. No discounts, coupons, support calls or manual concierge intervention anywhere in the flow.
OUT – SPECIFIC	No after-dispatch intervention. Every mechanism above acts before a vehicle is committed; a capacity flow at pickup was designed twice and rejected (§09). Existing screens are not restructured — the solution adds one flow, one conditional confirmation and one post-trip prompt, and touches nothing else. Repeat and business users are not served: they see the CTA, but it is not designed around them.

Release sequence

Scope is set by what each release can prove, not by what the design can hold.

RELEASE	SHIPS	WHY HERE
R0 • Instrumentation PRE - LAUNCH	The Stage 08 post-trip confirmation (5.8), to all first-time completed trips. No user-facing assistant.	Seeds the comparables corpus before anyone sees a recommendation. Without it the trust layer has nothing to show on day one, and the cold start becomes the launch's first failure rather than its solved problem.
R1 • The assistant COHORTS 0–3	Photo and text input (5.1–5.4), low-confidence fallback (5.5), Stage 04 load check (5.7).	Everything needed to test the core claim — that translating a physical job into a vehicle specification is the thing first-timers cannot do. Photo and text ship together because the fallback is the expensive shared component and gets built either way; splitting them would defer revenue without deferring cost.
R2 • Dimensional accuracy POST - GA	Longest-side gating wherever Porter publishes load-bay dimensions; a live city-catalogue availability feed.	Answers the two risks §15 names as most likely to produce a confidently wrong answer (R7, R8). Deferred rather than dropped: both need R1's accuracy data before they can be calibrated against anything.
R3 • Segment expansion GATED, NOT DATED	Load types beyond single-item and small-multi-item household goods, widened one type at a time.	Model accuracy tracks corpus density, so scope widens where the evidence loop has already run. This release has a trigger, not a date.

Open questions carried, not closed

Loading and unloading responsibility is an adjacent 3/3-convergence barrier this solution does not answer. It sits in the same pre-booking moment, on the screen immediately next door, and a vehicle-sizing recommendation does nothing for it. Idea #5 in §10 is the answer that was not chosen.

No confidence threshold for 5.5 or the 5.7 trigger — a Porter ML-team decision outside this document's authority. **The load check reduces mis-sized dispatches without eliminating them;** a load described inaccurately and sized confidently passes through unflagged, and §15 sizes that residual. **First-timer volume** (~15% of the individual base) is the softest input in the business impact model — an assumption, not a case-study given.

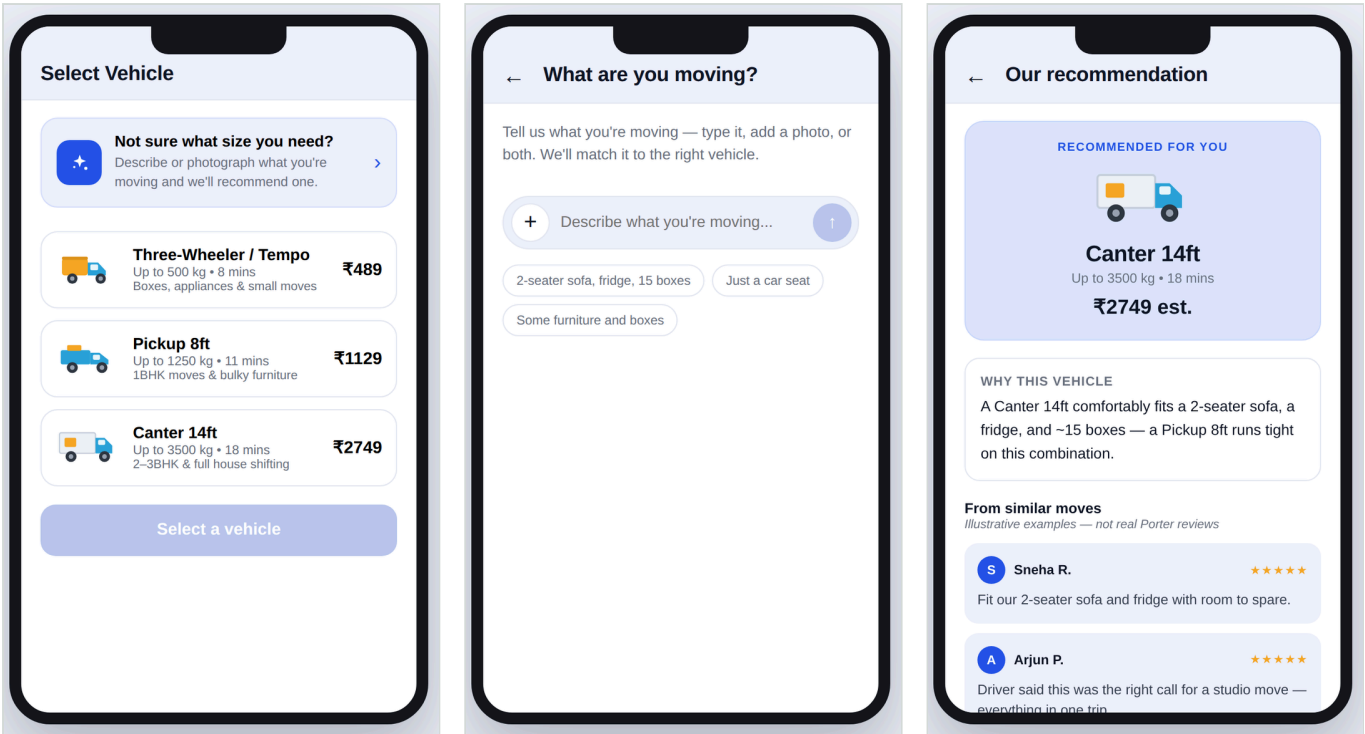
12

Prototype & annotated screens

STEP 4

Interactive prototype: <https://claude.ai/code/artifact/751e3234-cef2-4469-90c0-916a988222c2>

Six core screens plus the low-confidence fallback. Vehicle categories and capacities are Porter's own, verified on porter.in: Three-Wheeler 500 kg, Pickup 8ft 1250 kg, Canter 14ft 3500 kg. Fares shown are illustrative estimates for the demo route rather than Porter's published starting prices, which are lower. The demo route is a Delhi one because **Porter's catalogue varies by city** and Canter 14ft is offered in Delhi and Ahmedabad but not in Bangalore — see §15. Matching logic is deterministic keyword rules, tested against the full example set.



01 · Vehicle Selection MODIFIED

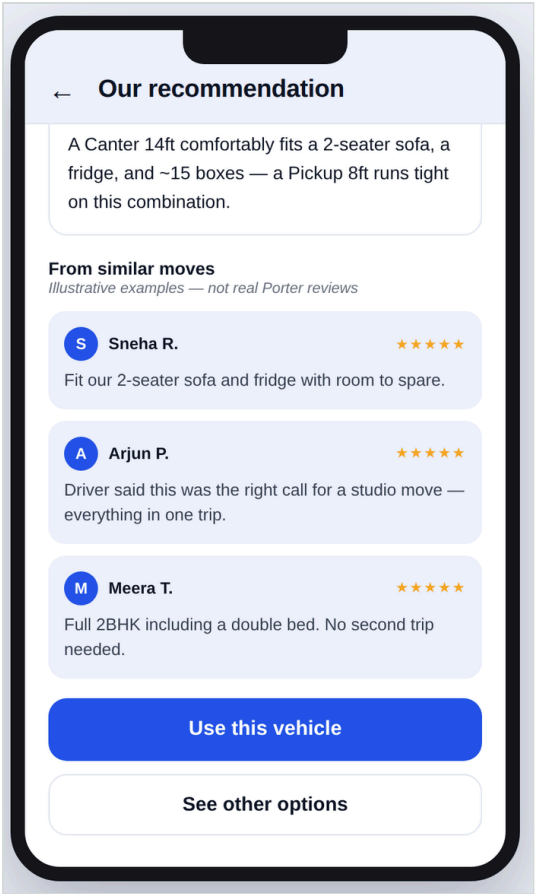
The existing category grid is untouched and remains the default. One new card sits above it — “*Not sure what size you need?*” Tapping a vehicle directly still works. Opt-in rather than a gate, because gating would force the new behaviour on repeat and business users who are out of scope, and would restructure a core existing screen.

02 · Assistant input NEW

One chat-style bar rather than separate photo and text modes — attach on the left, free text in the middle. Example chips lower the cost of the first input, which is the moment a first-timer is most likely to abandon a blank field.

03 · Recommendation NEW

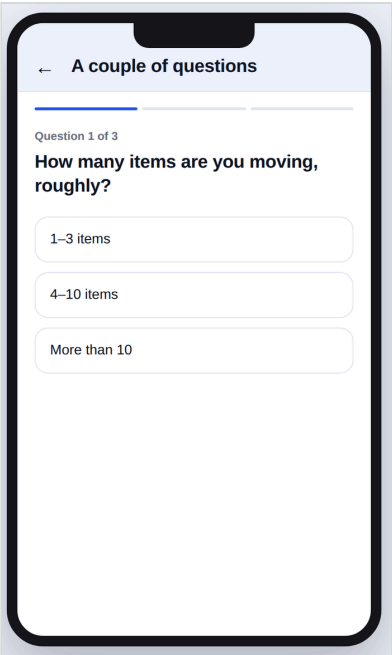
The screen the solution rests on. Vehicle, capacity, fare, and a *why* tied to the specific input: “*a Pickup 8ft runs tight on this combination.*” A vehicle name alone is another assertion a first-timer cannot evaluate — the reasoning is what makes it checkable.



04 · Trust layer NEW

Same screen, scrolled: comparable past moves in the same vehicle. R2 and R9 both gave *"didn't trust an app to send the right vehicle"* as their reason for going offline — trust and fit are one question to the user, so they are answered on one screen.

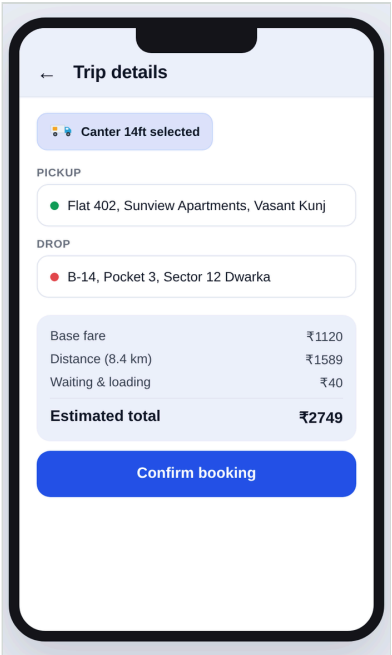
REVIEWS ILLUSTRATIVE — LABELLED ON-SCREEN



05 · Low-confidence fallback

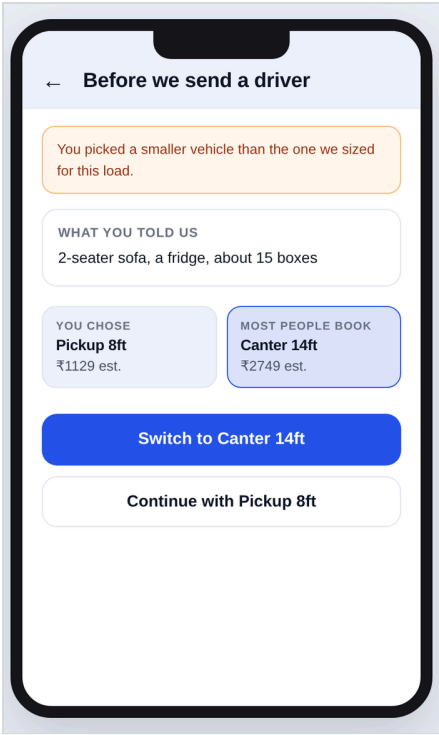
NEW

When the model cannot classify the load, it does not guess silently — it drops into two to three structured questions. This is the runner-up wizard's deterministic logic, absorbed as a fallback. No confidence threshold is specified: that is a Porter ML decision, not a number to invent here.



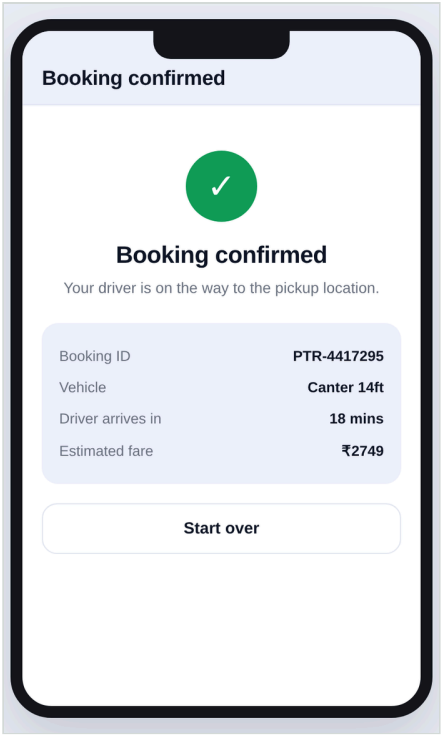
06 · Trip details EXISTING

Unchanged. The accepted recommendation pre-fills the vehicle field and the user rejoins the flow that already exists — visible here as the *Canter 14ft selected* chip above an otherwise standard fare breakdown.



07 · Load check NEW

Fires only on a flagged-uncertain booking — here a downward override, where the user picked a Pickup 8ft over the recommended Canter. Both options and both fares side by side, one tap either way. No driver is assigned yet, so a switch is a re-selection rather than a re-assignment.



08 · Booking confirmed EXISTING

Unchanged. Included because the case study asks for two to three existing screens alongside the new flow — the point being that everything between the two touchpoints is byte-for-byte the current experience.

The prototype ships with a reviewer-facing panel that tags each screen New or Existing as it is viewed, and offers four one-click demo paths — including the override that triggers the load check. Built directly rather than in a no-code tool; the interaction and matching logic are the deliverable, not the toolchain.

Two things the prototype states rather than hides

The comparable moves are generated examples and say so on screen. The review corpus contains no positive fit reports — a successful fit produces no review, the same structural silence that hides pre-booking abandoners. Presenting generated reviews as corpus evidence would be the fastest way to undermine the parts of this research that are real.

Matching is deterministic rules over the typed text, not a live model. The reasoning line and the recommendation both change with what is entered, which is what makes the interaction demonstrable; recommendation accuracy is a production ML question a prototype cannot evidence either way.

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Go-to-market, growth loop & rollout

STEP 5

The population this feature serves is defined by the North Star’s own denominator: first-time customers who have already reached the vehicle-selection screen. They are inside the funnel. This is an in-product launch, not a marketing one — the whole first-few-weeks question collapses into whether the CTA gets noticed.

Launch plan — a use-case segment, not a city

The case study offers either. The barrier is load-type-specific, not geographic: a first-timer in Bengaluru and one in Jaipur face the same failure to turn a sofa into a vehicle category. Geography is a rollout dimension here, not a targeting one — though \$15 adds a real geographic constraint, since Porter’s catalogue varies by city.

Launch segment: single-item and small-multi-item household goods moves. A refrigerator. A sofa. A washing machine. A room’s worth of cartons. Three independent reasons converge on it, and they are not of equal weight — the first is binding on its own.

SCOPE	Binding. Full home relocation is the Packers & Movers job, and a full Packers & Movers redesign is out of scope. Launching on relocations would put the first cohort outside the case study’s own boundary. The segment choice is not merely preferable — the alternative is disallowed.
TRACTABILITY	A bounded set of item types can be enumerated, which matters twice: model accuracy is highest where examples are densest, and the fallback’s rules table only has to cover a taxonomy that can actually be written down. Full relocations are combinatorially open — every one is a different list.
EVIDENCE	Directional. The loads named across the research base are single items and small household sets — Interviewee 1’s car seat, R5’s household move, R8’s unprompted request to scan “my things”. On 21 conversations that points rather than proves, and it is recorded as corroboration for a segment the two reasons above select on independent grounds.

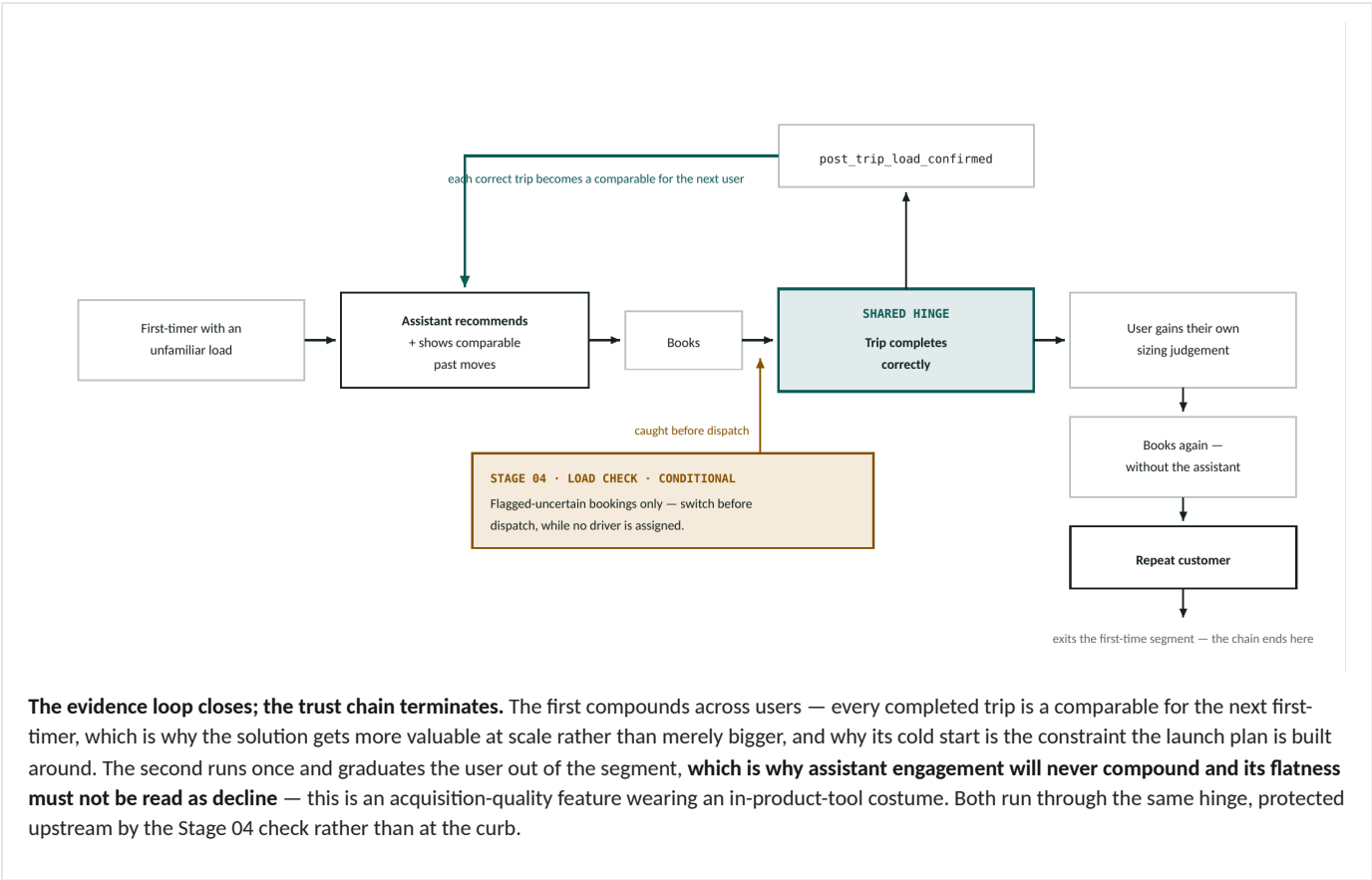
Adoption levers — all inside the scope guard

LEVER	WHAT IT IS	METRIC
CTA prominence	Placement, wording and visual weight of the Stage 01 entry point, A/B tested from Cohort 1. <code>sizing_cta_shown.cta_variant</code> exists for exactly this.	B1 — CTA engagement
First-time flagging	The CTA is prominent for zero-completed-trip individual accounts and merely present for everyone else. Targeting is an account property, not a campaign.	B1 by segment
First-session variant	A first-time account’s first arrival at Stage 01 is the highest-value impression this feature will ever get, and is treated as a distinct variant.	B1 by session
Return to an abandoned flow	A first-timer who previously reached Stage 01 without booking has the highest demonstrated need. Flagged, not committed — this is idea #8 in \$10, which did not make the top 2, and using it here would reintroduce a deprioritised idea through the side door.	B1, B5

The data dependency ships before the feature

The recommendation screen surfaces comparable past bookings. That corpus does not exist today, and it is what the trust half of the solution is made of. The post-trip prompt (PRD \$5.8) is independent of the assistant, so it ships to all first-time completed trips well ahead of any user exposure — and Cohort 1 opens against a populated corpus rather than an empty one. This is what makes the narrow launch segment do double duty: it concentrates corpus density on exactly the load types the first cohort will ask about.

Growth and behaviour loops



Cohort rollout

The six-month target sets the schedule. The case study fixes the 41% → 58% window at six months, so general availability has to land by month five to leave a measurable period at full exposure. That compresses the pilot to ten weeks — long enough for its earliest completions to enter their 90-day repeat window during Cohort 2, which is why G4 gates Cohort 3 rather than Cohort 2.

#	COHORT	SCOPE	GATE TO ADVANCE
0	Shadow 0% exposure WEEKS 1–3	The model runs silently on live first-time sessions; its recommendation is logged and compared against what the user chose and what actually completed. The post-trip prompt ships to all first-time trips to seed the comparables corpus.	F2 in shadow ≥ the completion rate of unaided manual selection. If the model is not better than a first-timer's guess, nothing downstream matters — and no user was exposed to find that out.
1	Pilot 5% WEEKS 4–13	2–3 metros chosen on first-timer volume and catalogue depth (\$15), launch segment only. Full flow live including the fallback, against a concurrent holdout. CTA wording A/B tested.	No guardrail regression vs. holdout (G1 G2 G3 G5). B1 above the discoverability floor. Absolute targets for B1–B7 are set from this cohort's own data — not invented before it.
2	Scale 25% WEEKS 14–20	Wider city set, same load segment. The evidence loop is now self-sustaining within the segment.	North Star directionally positive vs. holdout. Fallback exercised at enough volume to be trusted. F3 and F2b reading. G4 does not gate here — its 90-day window means Cohort 1's repeat signal is not yet readable.
3	General 100% WEEK 21 →	All Porter cities. Load-type scope widens one type at a time as corpus density permits — accuracy tracks density, so scope widens where the loop has already run, not where a roadmap says it should. A small holdout is retained permanently , or the North Star stops being attributable the moment the feature is everywhere.	North Star on a credible path to 58%. G4 read against holdout — the first point at which \$09's repeat-rate assumption is constrained by data rather than argued.

Kill and rollback criteria, stated in advance

So the decision is not made under pressure with the data already in front of us. **G3 regresses materially vs. holdout** → roll back one cohort; named in advance as the most likely trip, for the structural reason in §15. **F2 falls below the shadow-mode manual baseline** at any cohort → halt; the feature is making sizing worse than guessing. **G1 or G2 regress** → halt and diagnose via B4 override direction. **B1 high and the North Star flat** → not a rollback: the recommendation, not the CTA, is what failed, and the rollout pauses while the mechanism is re-examined.

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Monetization

The feature is not directly monetized. No fee, no tier, no upsell — an upsell inside a recommendation would corrode the exact thing the recommendation exists to establish, and pricing mechanics sit against the out-of-scope pricing-engine line.

The effect runs through **vehicle mix**, and its direction is ambiguous. Correcting a first-timer *upward* raises fare per trip and prevents a failed pickup. Correcting one *downward* lowers it — and that is not hypothetical: Interviewee 1 was pushed into a larger, costlier vehicle than the job needed. So the model holds AOV flat and treats mix shift as a **tracked outcome** — the distribution of `vehicle_selected.category` split by `source` — rather than a modelled gain. Assuming an uplift would be assuming the answer to the question the metric was built to ask.

One uncomfortable line, stated because a reviewer will find it anyway

Some of Porter's current revenue comes from mis-sizing: the upsize at the curb, the loading charge Interviewee 1 was hit with. A feature that gets the vehicle right the first time removes some of that revenue. The case for it is not that it protects every rupee of today's mix — it is that ₹1,400 of lifetime value per completed customer, at a 17-point higher completion rate, is worth more than the margin on a bad trip that ends a customer relationship at trip one.

Willingness to pay

Mix shift only matters if first-timers will pay the corrected fare, so willingness to pay is the commercial question underneath the whole design. A correct recommendation is frequently a *larger, costlier* vehicle.

The evidence says first-timers abandon on surprise, not on price. Interviewee 1 did not walk away from a high quote — the booking died on an unexpected loading charge after the fact. In \$07, *price-opacity* is a 3/3-convergence barrier; *price level* is not. Across 220 Porter reviews the complaint is that the final number differed from the estimate, not that the estimate was high.

THE QUESTION	THE INSTRUMENT	READS FROM
Will a first-timer accept a higher fare once they understand what the cheaper option costs them?	Metric B7 — the confirm-versus-switch split at the Stage 04 load check, which shows the user a fare difference and asks them to choose.	Cohort 1
Does a corrected fare read as a purchase or a penalty?	Guardrail G3 against holdout, with the F4 fare-delta distribution behind it.	Cohort 1
Which direction does the mix actually move?	<code>vehicle_selected.category</code> split by <code>source</code> , against holdout.	Cohort 1

This is a measurement, not an assumption. The Stage 04 load check is the only place in the flow where a first-timer is shown a price difference and asked to decide, which makes B7 a direct willingness-to-pay reading on the exact population and the exact moment that matter. Nothing in the business impact model depends on the answer — AOV is held flat precisely so the model does not front-run it.

14 Metrics & event tracking

STEP 6

North Star — First-Trip Completion Rate

The percentage of first-time individual customers who reach the vehicle-selection screen and reach completed-trip status within 7 days. **41% → 58% within 6 months** — both figures given by the case study.

DENOMINATOR	Distinct individual (non-MSME) accounts with zero prior completed trips that enter pickup and drop and view the vehicle-selection screen — Stage 1.
NUMERATOR	Of those, accounts reaching trip-completed status (Stage 8) within the window.
COMPANION	Completed first trips per month, reported beside the rate and never in place of it. Any rate improves if you shrink its denominator; if the rate rises and the volume does not, the rate moved for the wrong reason.

Supporting metrics — booking and fulfilment

ID	METRIC	WHAT IT TELLS YOU
B5	Stage 1→4 pass-through	The direct Leak 1 measure.
B1	Assistant CTA engagement	Discoverability. Opt-in caps the solution at its own discoverability; this is that ceiling, measured.
B2	Assistant completion rate	Whether the added step became new friction. A tool built to stop abandonment can create its own.
B3	Recommendation acceptance	Whether the recommendation is believed.
B4	Override direction	Larger, smaller or same tier. Separates users buying safety by upsizing from users buying price by downsizing — downsizing overrides are a leading indicator of Leak 2 failures.
B6	Time to booking, cohort level	The on-demand-speed constraint made measurable. A trade-off named in prioritization and never measured is a trade-off that was only ever rhetorical.
B7	Load-check switch rate	Of triggered Stage 04 load checks, the share ending in a vehicle change rather than a confirmation. A rate near zero means the triggers are firing on bookings that were already right and the step is pure friction; a high rate means the recommendation path is letting wrong vehicles through and the fix belongs upstream.
F1	Stage 4→8 pass-through	The direct Leak 2 measure. It moves through composition rather than through the fulfilment machinery — the assistant improves the sizing decision, and the load check intercepts the flagged-uncertain bookings before dispatch, so B5 rises sharply and F1 rises modestly. If F1 <i>falls</i> while B5 rises, the solution is pushing under-qualified bookings into fulfilment.
F2	Recommendation accuracy	Of accepted recommendations, the share completing with no vehicle change and no post-assignment cancellation. The only metric closing the loop between what the model said and what physically happened. This is the F2 the rollout in §13 gates on.
F2b	Override correctness	Of <i>overridden</i> recommendations, the share where the post-trip load record shows the original recommendation would have carried the load. No new instrumentation — it reads an event the corpus already collects. Without it, B4 gives the direction of overrides and not whether they were justified, and those two failures need opposite fixes.
F3	Vehicle change at pickup	Direct mis-sizing signal, read from two sources. A driver who settles a mismatch with cash at the curb logs nothing, so the dispatch-side event is a floor; the customer-side fit rating on the Stage 08 prompt is incentive-aligned and catches what the driver will not. The gap between the two is itself informative.
F4	Fare delta at completion	The consequence of mis-sizing the customer actually feels. Distribution, not mean.

Guardrails — all gated against a concurrent holdout

ID	GUARDRAIL	INSTRUMENT	GATE
G1	Cancellations	First-time post-booking cancellation rate, split by initiator and stage	No increase vs. holdout

ID	GUARDRAIL	INSTRUMENT	GATE
G2	Rebooking	Cancel-and-rebook within 24h, split by whether the vehicle category changed — a rebooking into a different category is a mis-sized recommendation confessing itself	No increase vs. holdout
G3	Fare disputes	Dispute rate on final fare, plus the F4 delta distribution	No increase vs. holdout
G4	Completed but unwilling to return	First-trip to second-booking conversion within 90 days, among first-timers whose first trip completed. Behavioural rather than survey-based: unbiased by response rate	No decrease vs. holdout
G5	On-demand speed	Median Stage 1 to booking-placed time for the <i>whole</i> cohort. Per-user the assistant necessarily takes longer — that is what it is for	No regression beyond the Cohort 1 threshold

G3 is the guardrail most likely to trip, for a structural reason

A correct recommendation is frequently a *larger* vehicle than a first-timer would have chosen, and a larger vehicle quotes a higher fare. Price-opacity is a 3/3-convergence barrier this solution does not address. The assistant can therefore be right about the vehicle and still widen the gap between what a user expected to pay and what they are quoted. That is a real and specific way this feature fails, and G3 is where it surfaces.

Event tracking specification

EVENT	FIRES WHEN	KEY PROPERTIES
vehicle_selection_viewed	Stage 1 renders	user_id , is_first_time , session_id , city , ts
sizing_cta_shown	CTA rendered	cta_variant , is_first_time
sizing_assistant_opened	CTA tapped	session_id
sizing_input_submitted	Photo and/or text submitted	input_type (photo / text / both), photo_count , text_length
sizing_fallback_triggered	Low-confidence path entered	reason (ambiguous_photo / sparse_text)
sizing_fallback_completed	Structured questions answered	questions_answered
sizing_recommendation_shown	Recommendation renders	recommended_category , confidence_bucket , reason_code , comparables_count , latency_ms
sizing_recommendation_accepted	User confirms	recommended_category
sizing_recommendation_overridden	User opens the manual grid instead	recommended_category , chosen_category , direction (up / down / lateral)
vehicle_selected	Category committed	category , source (assistant / manual_grid)
load_check_shown NEW	Stage 4 trigger fires	trigger_reason (low_confidence / downward_override / fallback_used), chosen_category , recommended_category
load_check_resolved NEW	User continues or switches	outcome (confirmed / switched), from_category , to_category
booking_placed	Stage 4	category , quoted_fare , source , load_check_shown
driver_assigned · pickup_started	Stages 5 and 6	booking_id
vehicle_change_requested NEW INSTRUMENTATION	Category changed at or after assignment	from_category , to_category , initiated_by (customer / driver)

EVENT	FIRES WHEN	KEY PROPERTIES
trip_completed	Stage 8	final_fare, fare_delta_vs_quote
trip_cancelled	Any stage	stage, initiated_by, reason_code
post_trip_load_confirmed	Post-trip prompt answered	actual_load_type, fit_rating (fit / tight / did_not_fit)
NEW rebooking_started	New booking within 24h of a cancellation	prior_booking_id, hours_since, category_changed

Why almost every target here is a gate rather than a number

The North Star has a number because the case study supplies one, and the volume companion has a number because it is arithmetic on that one. Everything else is a gate against a concurrent holdout. No usage data exists for a feature that has not shipped, so an absolute target for CTA engagement or recommendation acceptance would be a number invented to look decisive. A holdout gate needs no invented baseline, survives seasonality and everything else shipping in the same quarter, and is what actually governs a rollout decision. Absolute targets are set from the first cohort's own data.

Two diagnostics carry a decision rule instead of a target. B1 low and the North Star flat is a discoverability failure — the CTA is the problem. B1 high and the North Star flat is a mechanism failure — users find it, use it, and it changes nothing. Only the second says the recommendation is not the barrier's answer.

F2b crossed with B4 reads the same way for overrides. An **upward** override where F2b is high — the original recommendation would usually have fit — is a **trust problem**: the model, not the reasoning shown for it. An upward override where F2b is low is a **model problem**: the user was right to go bigger. A **downward** override where F2b is high is exactly the Leak 2 risk B4 exists to catch — a correct call overridden anyway.

15 Risks, trade-offs & constraints

STEP 7

A risk with no instrument is a worry, not a risk. Every row below names what would show it, in the metric vocabulary §14 already defines. The column that matters most is not severity — it is whether the funnel shows the problem *before* a customer experiences it.

Risk register — abridged; full version in the Step 7 document

#	RISK	SEV	VISIBLE FIRST?	INDICATOR	MITIGATION
R1	Nobody finds the CTA. Opt-in caps the solution at its own discoverability.	High	Yes	B1	CTA variant testing from Cohort 1; a distinct first-session variant. <i>Residual</i> : a gate would guarantee reach and break scope. Not available.
R5	Language. First-timers across 23 cities describe loads in Hindi, Hinglish and regional languages. A model tuned on English degrades <i>silently</i> — a confident recommendation from a partial parse, not an error.	High	Partly	fallback rate by city, F2	The fallback path is language-independent by construction — tappable structured options, no free text. Route into it aggressively below threshold. The strongest argument for keeping the fallback mandatory.
R6	A load described inaccurately and sized confidently. Passes all three Stage 04 triggers and fails at the curb.	High	No	F3, fit_rating	None in V1. The mitigation is measurement — the customer-side fit rating exists so this is counted rather than assumed. Its size decides whether V2 needs recovery.
R7	Dimensional mismatch. Porter's catalogue clusters five categories between 350 and 750 kg and publishes load-bay dimensions for only some. A sofa is a length problem wearing weight's clothing.	High	No	F2, F3	Make the oversized-item question mandatory whenever a single large item is named; gate on longest published dimension; round up where none is published.
R8	The recommended class is not offered in that city. Verified, not hypothesised.	High	Yes, if built for	F1 by city	Constrain the candidate set to the city's live catalogue <i>before</i> the model chooses. A filter, not a supply problem — fleet expansion is out of scope and is not the answer.
R9	Supply depth, not just supply. Recommending upward shifts demand toward thinner classes: longer waits and more driver cancellations for a <i>correct</i> recommendation.	Med	Partly	G1, F1 by category	Per-city supply depth becomes a cohort gate. Magnitude is Porter ops data this analysis does not have; stated as an assumption, not a figure.
R10	Driver-side renegotiation absorbs the failure invisibly. A mismatch settled in cash logs nothing.	Med	No	F3 source divergence	None available. Driver compensation is out of scope, which is precisely why the protection sits at Stage 04. The gap between F3's two sources is the measurement.

#	RISK	SEV	VISIBLE FIRST?	INDICATOR	MITIGATION
R12	Price shock. A correct recommendation is often a costlier vehicle, and the load check makes the delta explicit at the worst moment. Price-opacity is a 3/3 barrier this solution does not address.	High	Yes	G3, B5, B7	The only lever without touching pricing is language: name what the cheaper option costs — a second trip, or a load that does not fit — so the delta reads as a purchase rather than a penalty.
R15	The comparable moves are other customers' trips. Consent and anonymisation.	Med	Pre-launch gate	—	Opt-in at the post-trip prompt; first name and initial only; load type and fit only; no addresses, dates, or photographs without separate consent.
R17	First-time identification is account-level. Reinstalls, new accounts, shared logins.	Low	Yes	denominator drift	Accepted, not solved. Because the design is opt-in, the cost lands on <i>metric precision</i> , <i>not user experience</i> — the opt-in decision pays off a second time. The holdout keeps the comparison valid regardless.

Porter's catalogue is not the same in every city — verified on porter.in, 2026-08-29

Bangalore: E-loader 350 kg · Three-Wheeler 500 kg · Tata Ace 750 kg · Pickup 8ft · Tata 407. **Delhi and Ahmedabad:** the same, plus Canter 14ft 3500 kg. **National catalogue:** also Champion 650 kg, Eeco 500 kg, 8ft 1Ton 1000 kg.

The same load has a different correct answer in Bangalore than in Delhi, and a recommendation the local market cannot fulfil is worse than no recommendation — it is a specific promise the app cannot keep. **The recommendation engine needs a live availability feed, not a static category table**, and launch cohorts belong in markets whose catalogue is deep enough for the recommendation to have somewhere to go.

Trade-offs — what this design consciously does not optimise for

NOT OPTIMISED FOR	CHOSEN INSTEAD	WHAT IT BUYS	WHAT IT COSTS
Reach	Speed and scope compliance — opt-in, not a gate	Repeat and business users lose nothing; the feature stays inside the user boundary; engagement stays falsifiable	The ceiling equals discoverability. R1 is the direct consequence.
Recovery	Prevention — Stage 04, not Stage 06	A mechanism that will actually be used; no dispatch changes, no pricing changes	R6. A confidently mis-described load still fails at the curb.
The cheapest trip	The trip that completes	Fewer failed pickups and second trips; a completed first trip rather than a cheap failed one	R12. Systematically higher quoted fares on the loads the assistant touches.
Barrier breadth	Depth on one barrier	The primary barrier gets a real answer rather than four barriers getting gestures	Price-opacity, loading responsibility and support are all left untouched.
Determinism	Coverage — AI primary, wizard as fallback	Non-standard loads get an answer; the trust layer becomes possible	A failure mode a lookup table cannot have (R5, R7), and the most expensive build in the table.
Repeat and business users	First-time individuals	Matches the case study's boundary and the evidence base	The same tool would help a repeat user moving something unfamiliar. Deliberately not served.
Speed for the individual	Certainty for the cohort	More first trips complete	Every assistant user spends time a grid-tapper does not. G5 is measured at cohort level for this reason.

The trade-off a reviewer is most likely to press on

The design optimises for the trip completing, not for revenue per trip. That is defensible on the case study's own definition of success — a completed trip, not a placed booking — and on the lifetime-value arithmetic in §14's companion model, where ₹1,400 of completed-customer value at a 17-point higher completion rate outweighs the margin on a bad trip that ends the relationship at trip one. It is still a trade, and it is made knowingly.

Constraints shaping the first version

OPERATIONAL	No backend dispatch changes, so every intervention acts before a vehicle is committed — which is what put the load check at Stage 04. The fallback rules table must be maintained against a category line-up that varies by city and over time: ongoing cost, not a one-off build. Driver behaviour cannot be changed by this product.
PRICING	The fare engine is untouchable. The recommendation cannot discount, bundle, waive or lock a fare, and cannot soften the delta it surfaces. Discounts and coupons are banned levers, so the only tool against price shock is language.
CAPACITY	Fleet mix is given — fleet expansion and vehicle procurement are both out of scope. The recommendation can filter to what a city offers, but cannot create depth where a class is thin. Recommending upward is a demand-side intervention on fixed supply.
DATA	No post-trip load data exists today, so the comparables corpus starts empty — which is why the post-trip prompt ships ahead of the feature. No confidence threshold exists; that number is a Porter ML decision and is deliberately not invented. First-time identification is account-level and imperfect.
SCOPE	The out-of-scope list is enforced without exception. Three mitigations a reader might reasonably expect — a curb-side upgrade, a driver incentive to log mismatches, a fare adjustment on a corrected recommendation — are unavailable for exactly that reason, and are marked as such rather than quietly omitted.

Assumptions

Each with its basis and what would falsify it. Everything below is an assumption, not a finding.

ASSUMPTION	BASIS	WHAT WOULD FALSIFY IT
First-timers are ~15% of the ~10 lakh monthly individual base	Derived from the case study's 30 lakh monthly less 20 lakh MSME split; the <i>share</i> itself is illustrative	Porter's own first-time cohort counts. The softest input in the \$09 model — halve it and every rupee figure halves.
AOV ₹800, CAC ₹250, repeat rate 30%, 2.5 further trips over 24 months	Illustrative, carried consistently through every model in this pack	Actuals. G4 bounds the repeat-rate assumption from below at the end of Cohort 1 — a 90-day second-booking rate constrains a 24-month repeat rate rather than measuring it.
Larger vehicle classes have thinner supply than three-wheelers and Tata Aces	Inference from Porter's positioning as a mini-truck aggregator, and from Canter 14ft's absence from the Bangalore catalogue. Not verified — no public supply-mix figure exists	Porter ops data. R9's magnitude depends entirely on this.
A first-timer who gets the right vehicle is more likely to book again	Argued in the trust chain (§13); deliberately not built into the base case	G4 against holdout. Worth ₹10.4 Cr per annual cohort in the sensitivity if true.
Most first-time loads fall inside a tractable taxonomy of household items	The loads named across the research base are single items and small household sets; on 21 conversations this is directional , not conclusive	The fallback trigger rate. If it fires constantly, the taxonomy is too narrow and the launch segment is wrong.
Shadow-mode accuracy predicts live accuracy	Standard shadow-mode assumption; users behave differently when a recommendation is visible to them	Cohort 1's F2 diverging from Cohort 0's.

Three pre-launch gates that are not metrics

A **city-catalogue availability feed** (R8), a **comparables consent and anonymisation policy** (R15), and a **legal read on recommendation wording** (R16) — “our recommendation” and “most people moving this list book...”, never “guaranteed to fit”. None of the three is measurable, and all three must clear before Cohort 1 opens.